New Kernel Functions can be derived From other Kernels

Sum of Kernels is a Kernel

$K(a,b) = K_1(a,b) + K_2(a,b)$

Scalar Product of Kernel is a Kernel

$K(a,b) = \alpha K_1(a,b)$

Product of Kernels is a Kernel

$K(a,b) = K_1(a,b) * K_2(a,b)$

Combination of the above rules is a Kernel

$K(a,b) = K_1(a,b) * K_2(a,b) + \alpha K_3(a,b)$